Canada's Population Trends

Population is defined as the number of people occupying a given area. Population statistics are collected every five years in Canada by a national Census. Census takers used to go door-to-door and ask each household a variety of questions related to the number of persons in a household, age, occupation, family income, ethnic background and so forth. In more recent years, census questionnaires are sent out by mail in most areas that are accessible by postal service. Statistics questions have increased to include counts on possessions such as number of appliances, cars, electronic equipment and more. Other data collected are statistics on commercial trade import and export and transportation. Farm statistics collect information from the number of animals raised to the amount spent on fertilizer.

Although it may seem invasive for the government to question you on your personal information, census information is legislated by government to protect your personal privacy. Not even Revenue Canada or the RCMP can access individual information. Names and addresses are collected only to insure that all persons and households are included and not entered more than once. This specific information is not included in the database. Data that is collected is grouped and averaged for a census plot of many individuals so that information becomes representative and not specific.

The collected data is important to the government for: planning social programs such as Old Age Security and Child Tax Benefits; municipal services to establish schools and day-care centres; determining the extent of social and physical disabilities within the population; determining migration patterns; planning post-secondary and adult education programs; developing job creation programs; identifying specific geographic areas that need social assistance; determining areas that require housing programs.

Once the data is amalgamated to a minimal census plot, some of the information is made available to the public. Population is one of the databases that is accessible by all Canadians. We shall focus on the personal aspect of the population of Canada, i.e., number of people, dwellings, income, education, and so forth.

Population numbers tell us how many people are in a given area, but not all areas are equal in size. To make figures comparable, a population density is calculated which tells us the average number of people in a square kilometer. This allows us to compare population data at different levels by grouping the information collected in the census plots within the area of a municipality, province or ecozone respectively. If we look at Table 1, we see that densities vary greatly across the ecozones and show variations between Census years 1971 through 1991. Let's look at these five Census years and find a *population trend*, that is, the direction in which an ecozone population is going: increasing, decreasing or remaining relatively constant.

Table 1: Density Table for five Census years for each Ecozone

Ecozone names	Census 1971	Census 1976	Census 1981	Census 1986	Census 1991
Arctic Cordillera	0.00654	0.00852	0.00636	0.00717	0.00812
Northern Arctic	0.00830	0.00928	0.01036	0.01237	0.01425
Southern Arctic	0.00598	0.00847	0.01142	0.01312	0.01447
Taiga Plains	0.02725	0.03163	0.03247	0.03608	0.03790
Taiga Shield	0.02122	0.02228	0.02463	0.02179	0.02724
Boreal Shield	1.45452	1.53008	1.56236	1.36554	1.61978
Atlantic Maritime	11.39533	11.82850	12.17778	12.40204	12.60761
Mixedwood Plains	99.19291	105.30327	109.51450	114.93109	126.05242
Boreal Plains	0.83814	0.91238	0.98912	1.01917	1.04216
Prairies	6.40050	6.02559	7.68121	8.07801	8.44914
Taiga Cordillera	0.00086	0.00089	0.00224	0.00212	0.00123
Boreal Cordillera	0.04454	0.05329	0.05728	0.05723	0.06664
Pacific Maritime	8.30110	9.16502	10.11289	10.83429	12.57036
Montane Cordillera	1.06337	1.29732	1.46408	1.45934	1.31850
Hudson Plains	0.02719	0.02528	0.02425	0.01721	0.02689

Population data can be assessed in different ways. Population density was compared above. From Table 2 below, we will learn how density values are calculated and how ratios can be used to show us information we might have missed.

Table 2:How do we calculate densities and ratios?

Ecozone Names	Land Area (km2)	Population 1996	No. of dwellings 1996	Density (pop./area)	No. persons/ dwelling
Arctic Cordillera	236309	1196	266	?	?
Northern Arctic	1338353	18881	4825	?	?
Southern Arctic	645355	11729	2892	?	?
Taiga Plains	549116	23986	7632	?	?
Taiga Shield	1077604	36560	10565	?	?
Boreal Shield	1620871	2894961	1082951	?	?
Atlantic Maritime	196691	2549061	963871	?	?
Mixedwood Plains	109779	14840411	5626811	?	?
Boreal Plains	664274	745172	259095	?	?
Prairies	454046	3979522	1499905	?	?
Taiga Cordillera	260373	358	128	?	?
Boreal Cordillera	450921	30324	11386	?	?
Pacific Maritime	201670	2848289	1098179	?	?
Montane Cordillera	471117	851656	326977	?	?
Hudson Plains	346016	11811	2902	?	?

To find the population density, for example in 1996, we will look at the number of people per square kilometer in each ecozone. The population density is calculated by dividing the population of each ecozone by its area in square kilometers.

Now, let's use a ratio to find the average number of people in each dwelling. To do this we will divide the population by the number of dwellings. (Dwelling is used to indicate a person's home, be it a single house, an apartment or any other accommodation).

Table 3: CENSUS Data for Populations of Provincial and Territorial Capitals in Canada

City	Population 1996	Area km^2 1996	Density per km^2
Vancouver, BC	1 831 665	2820.66	649.3746
Edmonton, AB	862 597	9536.63	90.4509
Regina, SK	193 652	3421.58	56.5972
Winnipeg, MB	667 209	4077.64	163.6263
Toronto, ON	4 263 757	5867.73	726.6451
Quebec City, QC	671 889	3149.65	213.3218
Moncton, NB	59 313	142.37	416.6116
Halifax, NS	332 518	2503.10	132.8425
Charlottetown, PE	32 531	42.64	762.9221
St.John's, NF	174 051	789.66	220.4126
Iqaluit, NU	4 220	45.06	93.6529
Yellowknife, NT	17 275	102.38	168.7341
Whitehorse, YK	19 157	413.48	46.3311

Note: populations are for the CENSUS metropolitan areas (e.g.Toronto is the greater area only)

GEOMATICS
FOR HIGH SCHOOL CURRICULUM